D03 · Co-Manufacturing Operations

02 · Technology transfer to contract partners

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1 WHAT IT IS

Technology transfer to contract partners is the structured process of moving a manufacturing capability from a source site (typically an internal site or a previous partner) to a receiving site (the new contract partner), encompassing both the explicit knowledge (specifications, bills of materials, standard operating procedures, equipment requirements, process parameters) and the tacit knowledge (operator-level adjustments, process intuition, failure-mode patterns, supplier and raw-material handling specifics) that the source site has accumulated. The transfer is successful when the receiving site produces the same product at the same quality and yield as the source; the transfer is incomplete when documentation is complete but performance is not.

2 WHY IT MATTERS

Technology transfer is the most consequential single event in any co-manufacturing relationship — and the activity most consistently underestimated in scope, time, and organizational complexity. **The dominant failure mode is technology transfer executed as a documentation delivery: BOMs, specifications, SOPs, equipment lists, and process parameters are packaged and handed to the receiving site, the receiving site produces following the documentation, and the outputs do not match the source site because the tacit knowledge — the eighteen months of process adjustments, the operator-level recognition of equipment drift, the cleaning procedure that nobody wrote down — never transferred**. The canonical peer-reviewed work on technology transfer between manufacturing sites frames this gap directly **(Galbraith, 1990 — peer-reviewed, "Transferring Core Manufacturing Technologies in High-Tech Firms", California Management Review)**: manufacturing technology is composed of explicit knowledge that documents capture and tacit knowledge that documents cannot; effective transfer requires both, and most transfer programs systematically under-invest in tacit-knowledge capture and transfer. Industry data confirms the asymmetry — **technology transfers that include sustained source-site engineer residency at the receiving site, structured tacit-knowledge capture, and parallel production for validation produce receiving-site yields within 3–8% of source-site benchmarks within 6–12 months; transfers executed as documentation deliveries produce receiving-site yields 20–40% below source-site benchmarks even after 18+ months**. The gap is not technical; it is methodological.

3 HOW IT IS DONE

1. **Capture tacit knowledge from the source site before starting the transfer — not after the receiving site struggles.** Tacit-knowledge capture techniques: structured operator interviews (what adjustments do you make when X happens?), shadow-and-document sessions (engineer follows operator and documents undocumented decisions), failure-mode reviews (what's gone wrong historically and how was it diagnosed?), tribal-knowledge inventories (what does each operator and engineer know that's not in any document?). The capture step should add 4–8 weeks to the transfer timeline and produce a structured tacit-knowledge document that becomes part of the transfer package, not a separate deliverable.
2. **Embed source-site engineers at the receiving site for the first 6–12 production runs — not just for handover training.** The single most predictive factor for transfer success is sustained source-site engineering presence at the receiving site through the first batches. Source-site engineers detect when receiving-site behavior is producing different results before measurements catch it, transfer the tacit decision-making in real-time as situations arise, and build trust with receiving-site operators that documentation alone cannot establish. Transfers that include 3+ month source-site residency consistently succeed; transfers that include only handover training consistently struggle.
3. **Run parallel production at source and receiving sites with explicit comparison of inputs, outputs, and yields — not just at the end of transfer but throughout.** Parallel production means the same product is being produced at both sites with the same inputs and measured outputs, and weekly comparison shows where the receiving site is converging or diverging from the source. Differences become the working agenda: input variation? operator technique difference? equipment state difference? process-parameter divergence? Parallel production with structured comparison is the diagnostic apparatus that produces convergence; serial production (source stops when receiving starts) destroys the comparison signal and produces no diagnosis when divergence occurs.

4. **Validate the receiving-site equipment as a separate step — equipment of the "same model" is not necessarily the same equipment.** Two ostensibly identical machines from the same manufacturer can produce different results depending on age, calibration state, maintenance history, installed accessories, environmental conditions, and operator usage patterns. The equipment-validation step measures equipment-level outputs against source-site equipment with the same inputs, characterizes any differences, and either remediates the equipment or adjusts the process parameters for the receiving-site equipment specifically. Transfers that assume "same model = same equipment" routinely encounter equipment-driven yield gaps that look like process problems.
5. **Define explicit acceptance criteria for transfer completion — not vague convergence statements.** Acceptance criteria should be measurable, time-bounded, and gateable: receiving-site yield within X% of source-site benchmark, first-pass quality at Y level or better, cycle time within Z%, three consecutive production runs meeting all criteria. Transfer is "complete" only when criteria are met; if not, the transfer continues with active engineering support. Transfers without explicit acceptance criteria drift into ambiguous "ongoing improvement" phases where neither party knows whether the transfer succeeded.
6. **Plan and budget for the 6–18 month convergence period — transfers don't end at production launch, they end at performance parity.** Most organizations close out tech transfer at production launch (first commercial batch produced at receiving site); the actual transfer typically runs 6–18 months longer until receiving-site performance reaches source-site benchmarks. Budgeting for this extended convergence period — sustained engineering support, parallel production, ongoing tacit-knowledge transfer — is what separates transfers that converge from transfers that drift. Operations that under-budget the convergence accept persistent underperformance and call it the cost of contract manufacturing; operations that budget honestly produce convergence and capture the cost benefit the relationship was supposed to deliver.

WORKED EXAMPLE — ILLUSTRATIVE

A consumer electronics company transfers production of an established mid-volume product (45K units/month) from its internal Site A to a new contract partner Site B to free internal capacity for new products. **Initial transfer (documentation-heavy): comprehensive package handed over including all SOPs, BOMs, process parameters, equipment specs, and quality criteria. Site B receives 1-week handover training at Site A; production launches at month 4**. Performance at month 6: **Site B first-pass yield 67% vs Site A baseline 94%, defect signatures different from Site A's (Site B sees more cosmetic defects; Site A sees more functional defects), Site B cycle time 28% longer than Site A's, customer complaints traceable to Site B units running 4x the Site A historical rate**. Root cause analysis reveals: 18 months of tacit process knowledge at Site A — operator-level adjustments during humidity changes, specific cleaning procedures between SKU changeovers, equipment-drift warning signs that operators had learned to recognize, supplier-specific raw-material handling — had never been documented and never transferred. The restructured transfer (executed on the next product): **(1) 6-week structured tacit-knowledge capture at Site A before transfer began (operator interviews, shadow sessions, failure-mode reviews); (2) 3-month residency program with Site A engineer at Site B for first batches; (3) parallel production at Site A and Site B for 4 months with weekly comparison of inputs, outputs, yields; (4) equipment-level validation comparing Site B equipment against Site A reference; (5) explicit acceptance criteria gating transfer completion: 90% yield, defect signature matching Site A, cycle time within 10%**. Outcome: **Site B yields converge to 91% within 8 months (vs persistent gap on the previous transfer), defect signatures match Site A's, cycle time within 6%, transfer "complete" gate passed at month 9**. The economic comparison: **extra transfer cost ~$340K in engineering time and parallel production; saved cost from yield gap, quality issues, and customer complaints on first 24 months ~$4.2M**. The fast transfer was the expensive one in production output, customer trust, and source-site engineering hours spent firefighting.

IN PRACTICE

Capture tacit knowledge from the source site before the transfer begins

The operator-level adjustments to humidity changes, the engineering judgment about which precursor sign is real trouble, the cleaning procedure nobody bothered to document **(Galbraith, 1990 — peer-reviewed, California Management Review)**. This capture is the structural innovation that distinguishes transfers that converge from transfers that drift.

Embed source-site engineers at the receiving site for 6–12 production runs

Not just for handover training. The tacit knowledge transfers in the production environment under real conditions — not in training sessions. The residency cost is real and visible; the absence-of-residency cost is invisible until yield gaps materialize 6 months in.

Run parallel production with weekly comparison of inputs/outputs/yields

The source site and receiving site produce the same product against the same specifications for an overlapping window of 8–24 weeks. Weekly comparison surfaces drift while remediation is cheap; the alternative — sequential transfer with launch at the receiving site — surfaces the same drift after thousands of units have shipped.

Validate receiving-site equipment as a separate step — same model != same equipment

Equipment wears differently, is maintained differently, has different installation tolerances. A piece of equipment with the same model number at the receiving site is not the same equipment as the source site — calibration, characterization, and capability assessment for the specific equipment is a distinct transfer step.

WHAT YOU WOULD USE

TECHNOLOGY TRANSFER METHODOLOGY

MasterControl Quality Excellence / Veeva Vault QualityDocs / Sparta Systems TrackWise — tech-transfer document and process management — Enterprise platforms with structured tech-transfer workflow, document control, parameter versioning, and validation tracking; the right fit for regulated industries (pharma, medical device, food) where tech transfer has formal regulatory documentation requirements **(MasterControl — vendor; Veeva — vendor; Honeywell/Sparta — vendor)**.

Siemens Teamcenter / PTC Windchill / Dassault ENOVIA — PLM platforms with tech-transfer modules — Product lifecycle management platforms that maintain the master engineering record and support controlled tech transfer to manufacturing partners through structured BOM, specification, and process-parameter handoff **(Siemens — vendor; PTC — vendor; Dassault Systèmes — vendor)**.

External tech-transfer consultancies and specialized providers (Lonza, Catalent, Thermo Fisher consulting practices) — Specialized service providers with tech-transfer expertise across regulated industries; the right fit for operations executing major transfers without sufficient internal methodology depth, particularly in pharma and biotech where the regulatory complexity is highest **(Lonza — vendor; Catalent — vendor; Thermo Fisher — vendor)**.

A structured tacit-knowledge capture methodology + source-site engineer residency + parallel production discipline + explicit acceptance criteria — For operations without platform investment — a documented tacit-knowledge capture process (operator interviews, shadow sessions, failure-mode reviews), engineering staffing for source-site residency, parallel production budget allocated, acceptance criteria gating transfer closeout. The lowest-tech option produces the strongest outcomes when the methodology rigor is real and the schedule discipline is enforced.

4 SO WHAT

Technology transfer that delivers documents but not tacit knowledge produces a partner that follows specifications literally and does not replicate results — because the specifications were never the complete recipe. Capture tacit knowledge from the source site before the transfer begins, embed source-site engineers at the receiving site for 6–12 production runs not just for handover training, run parallel production with weekly comparison of inputs/outputs/yields throughout transfer, validate receiving-site equipment as a separate step because same model != same equipment, define explicit measurable acceptance criteria that gate transfer completion, and budget honestly for the 6–18 month convergence period after production launch. The 20–40% yield gap between documentation-only transfers and structured transfers with residency, parallel production, and tacit-knowledge capture is not technical — it lives entirely in whether the methodology recognized tacit knowledge as an asset to be transferred or treated only explicit knowledge as the deliverable.

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